

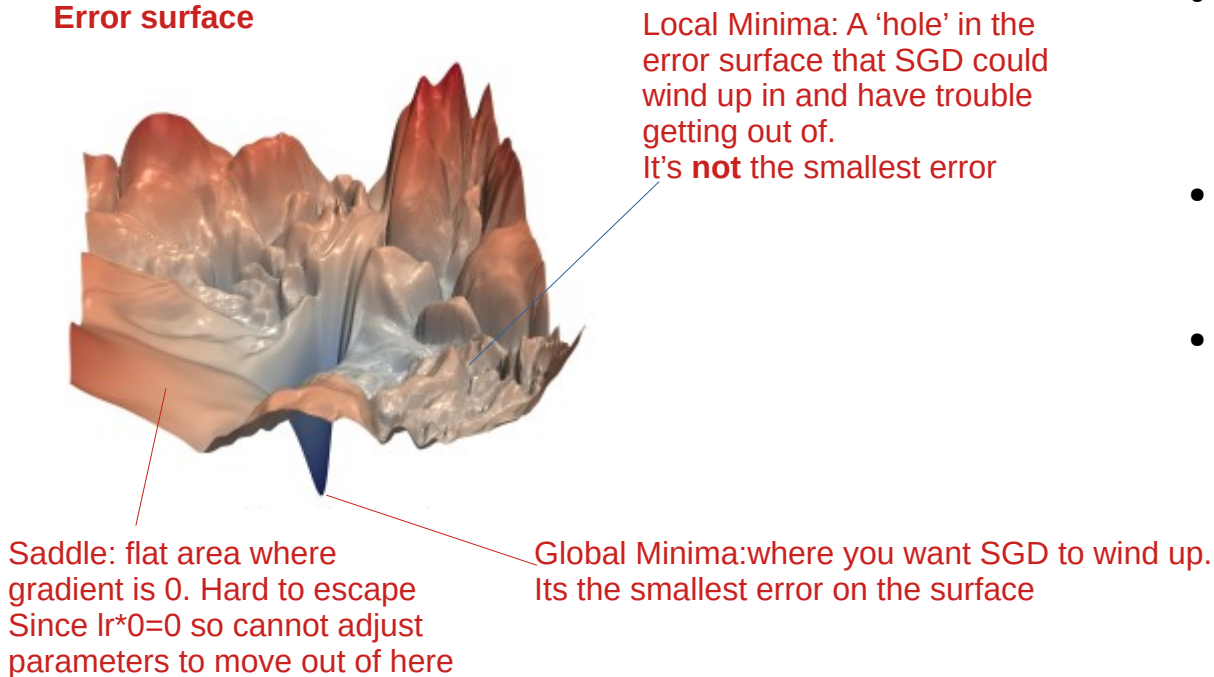
Optimizers- Gradient Descent (GD) verses Adam

Outline

- Error surface
 - Local Minima, saddle, global minima
- Optimizers
 - Gradient Descent
 - Adam
- Tradeoffs
- Further Reading

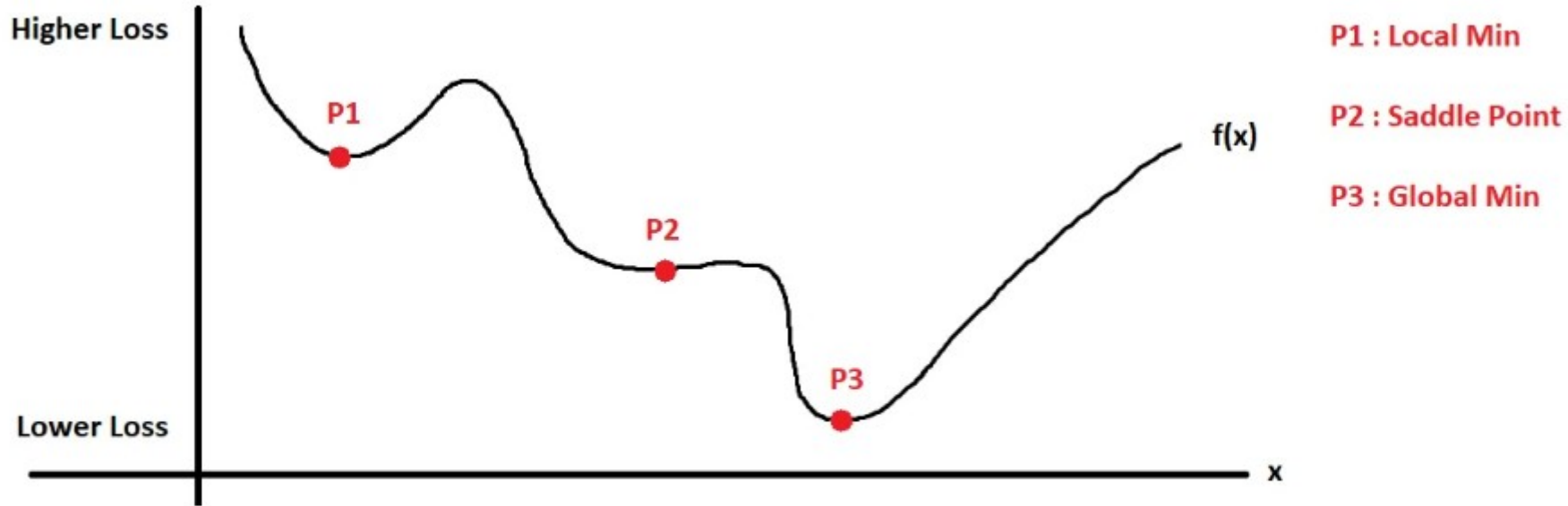
Error Surface

Error surface



- Projected error surface of a Resnet 54 CNN for 1 batch of data
- Millions of parameters projected into 3d space
- Want to get the error to the global minima, but might get stuck in a saddle or local minima with GD

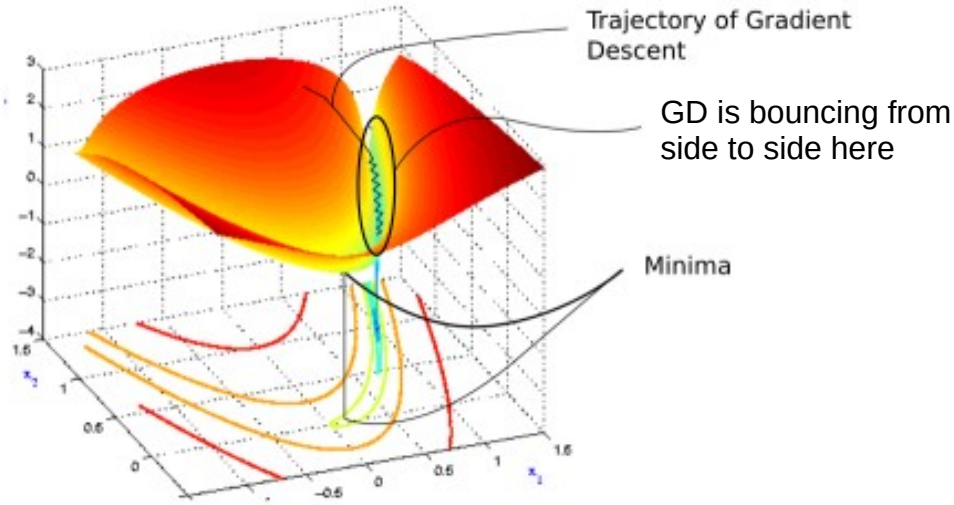
Error Surface



Optimizer: Gradient Descent

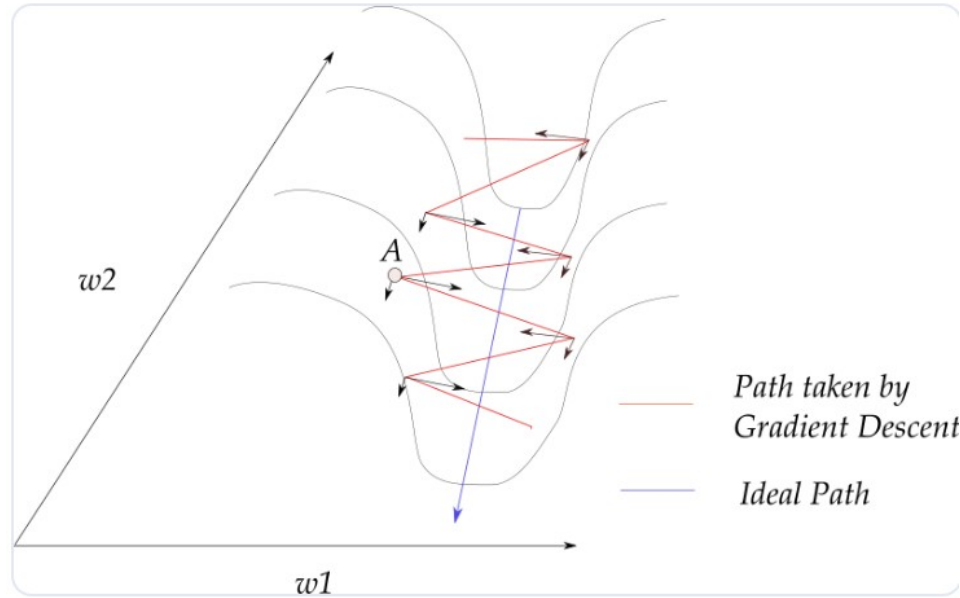
- GD analogy:
 - Analogy: A person taking equal-sized steps straight downhill
 - Updates the model params to reduce error
 - You get there but might get stuck in local minima or a saddle
 - Mini batch GD helps
 - every batch has different error surface
 - Even if stuck for this batch (no param update) will probably not be stuck for the next batch (param update)
 - See **Gradient Descent**

Optimizer: Gradient Descent



GD in a narrow 'canyon' will overshoot when params are updated and will 'bounce' from side to side

Optimizer: Gradient Descent with Momentum



- Movements decomposed into 2 directions
- If we add these vectors, the zig zag ones cancel out
- Left with vectors pointing in correct direction

Optimizer: Adam-momentum

- Momentum using an exponential decay
- a fraction of previous updates added to current update
- Increases update in direction of descent
- Helps to smooth noisy gradients and get thru small dips and saddles
- **Analogy: Pushing the person walking down the hill in the right direction**

Optimizer: Adam-adaptive step size

- If high gradient variance (big curvature in error surface) then shorten step size
- If not, lengthen step size
- Effect: Take smaller steps in steep/unstable directions and bigger steps in gentle/flat directions, reducing zig-zag in ravines

Optimizer: Adam- Why its popular

- Stability – out of the box params work well
- Faster early progress: Momentum + adaptivity helps escape plateaus/saddles quickly.
- Handles ill-conditioning: In skinny-steep valleys, Adam naturally dampens the steep direction and pushes along the flat one.

Adam: Tradeoffs

- Generalization: On some vision tasks, SGD + momentum + a good LR schedule can match or even beat Adam's final test accuracy.
- Weight decay: Use AdamW (decoupled weight decay) rather than Adam with L2; it regularizes more cleanly.
- Not a silver bullet: Truly dead ReLUs, terrible data scaling, or wild learning rates will still cause trouble.

Other Optimizers

- AdamW- has weight decay
- RMSProp – Adam derives partly from this
- Adagrad
- Adadelta

Further Reading

For further detail see;

Equations behind GD, Momentum and Adam

Optimizer Visualizers